

A Complemented Sublattice Need Not Inherit Banach-Lattice Semiprojectivity

Run fa_banach_001 — lane 17

11 August 2026

Status: counterexample, subject to human review. In the standard bounded-linear-complementation sense, semiprojectivity is not inherited by complemented sublattices: $C[0, 1]$ is semiprojective and contains a 2-complemented lattice-isometric copy of the non-semiprojective lattice c_0 .

1 Source problem and scope

Kania and Niwiński introduce Banach-lattice semiprojectivity in arXiv:2604.10624v1 [1]. Their Question 5.14 on arXiv PDF page 16 asks:

Source Question 5.14 (relevant part). *Which permanence properties—including passage to complemented sublattices, quotients by closed ideals, and Fremlin tensor products—does semiprojectivity enjoy?*

The full question also mentions finite ℓ_1 - and ℓ_∞ -sums. The present result answers only the complemented-sublattice branch.

Proposition 5.10 settles the finite-dimensional case, while Corollary 5.7 and Corollary 5.12 rule out the classical sequence lattices ℓ_p ($1 < p < \infty$) and the function lattices $L_p([0, 1])$ ($1 \leq p < \infty$). Proposition 5.13 extends [11, Example 3.3(iii)]. Recall also that $H^\Phi(\mu)$ is the order-continuous part of $L^\Phi(\mu)$ by [12, Proposition 4.7]; thus the additional hypothesis $L^\Phi(\mu) = H^\Phi(\mu)$ is precisely the order-continuity of the Luxemburg norm.

Question 5.14. Is the class of semiprojective Banach lattices closed under suitable finite direct sums (e.g., $\oplus_1$ or $\oplus_\infty$)? More generally, which permanence properties (passage to complemented sublattices, quotients by closed ideals, Fremlin tensor products) does semiprojectivity enjoy? As a basic permanence property, semiprojectivity passes to contractive retracts (Proposition 2.6).

Remark 5.15. Even the case of the ℓ_1 -sum $A \oplus_1 B$ (or ℓ_∞ -sum $A \oplus_\infty B$) appears subtle:

Here a *complemented sublattice* means a closed vector sublattice which is the range of a bounded linear projection. This is the standard Banach-space complementation meaning; no positivity or lattice-homomorphism property is imposed on the projection.

2 Idea of the proof

Use disjoint triangular bumps supported on intervals that shrink toward the origin. Their closed span is exactly a lattice-isometric copy of c_0 . Ordinary point evaluations at the bump peaks

would produce a sequence converging to $f(0)$, not to zero. Subtracting the value at a nearby endpoint fixes this: uniform continuity makes the difference coefficients tend to zero, while each difference still extracts the corresponding bump coefficient. This gives an explicit norm-two projection.

The source paper supplies both semiprojectivity verdicts needed to finish: Theorem A says $C(X)$ is semiprojective exactly when a compact metric X is an absolute neighbourhood retract, and Corollary 5.8 says that c_0 is not semiprojective.

3 Counterexample

Theorem 1. *There exist a semiprojective Banach lattice E and a closed sublattice $Y \subseteq E$ such that:*

1. Y is lattice-isometric to c_0 ;
2. there is a bounded linear projection $P : E \rightarrow Y$ with $\|P\| = 2$; and
3. Y is not semiprojective.

Consequently, semiprojectivity does not pass to complemented sublattices in the bounded-linear-complementation sense.

Proof. Set $E = C[0, 1]$. For $n \geq 1$, let

$$a_n = 2^{-2n}, \quad b_n = 2^{-2n+1}, \quad t_n = \frac{a_n + b_n}{2}.$$

The intervals $[a_n, b_n]$ are pairwise disjoint and converge to 0. Let $u_n \in C[0, 1]$ be the triangular function supported on $[a_n, b_n]$, affine on $[a_n, t_n]$ and $[t_n, b_n]$, and satisfying

$$0 \leq u_n \leq 1, \quad u_n(t_n) = 1, \quad u_n(a_n) = u_n(b_n) = 0.$$

For $\alpha = (\alpha_n) \in c_0$, define

$$J\alpha = \sum_{n=1}^{\infty} \alpha_n u_n.$$

Because the supports are disjoint and $\alpha_n \rightarrow 0$, this is a continuous function (also at the only accumulation point 0), and

$$\|J\alpha\|_{\infty} = \sup_n |\alpha_n|.$$

Since every u_n is positive and the supports are disjoint, J preserves absolute values and hence all lattice operations. Thus J is a lattice isometry onto the closed sublattice

$$Y := \overline{\text{span}}\{u_n : n \geq 1\} \subseteq C[0, 1].$$

Define $T : C[0, 1] \rightarrow c_0$ by

$$(Tf)_n = f(t_n) - f(a_n).$$

Both t_n and a_n tend to 0, so continuity of f gives $(Tf)_n \rightarrow 0$. Moreover,

$$\|Tf\|_{\infty} \leq 2\|f\|_{\infty},$$

so T is bounded. For every $\alpha \in c_0$, disjointness of the supports and the endpoint conditions give

$$(TJ\alpha)_n = (J\alpha)(t_n) - (J\alpha)(a_n) = \alpha_n.$$

Therefore $TJ = \text{id}_{c_0}$ and

$$P := JT : C[0, 1] \longrightarrow Y$$

is a bounded projection with $\|P\| \leq 2$. The norm is exactly 2: fix one n and choose $f \in C[0, 1]$ with $\|f\|_\infty = 1$, $f(t_n) = 1$, and $f(a_n) = -1$. Then $|(Pf)(t_n)| = 2$.

The interval $[0, 1]$ is a compact metric absolute neighbourhood retract, so Theorem A of [1] makes $E = C[0, 1]$ semiprojective. Corollary 5.8 of the same paper (whose statement is on arXiv PDF page 12 and proof on page 13) says that c_0 is not semiprojective. Since Y is lattice-isometric to c_0 , Y is not semiprojective. Hence Y is the required complemented sublattice of the semiprojective lattice E . $\square$

4 Verification and limitations

The only convergence point requiring attention is 0. It is harmless for $J\alpha$ because $\alpha_n \rightarrow 0$, and for Tf because $f(t_n) - f(a_n) \rightarrow f(0) - f(0) = 0$. The identities $TJ = \text{id}_{c_0}$ and $P^2 = P$ are exact, not approximate. The proof is otherwise elementary once Theorem A and Corollary 5.8 of the source are accepted.

The projection P is not claimed positive or lattice-homomorphic. Thus the theorem does not contradict Proposition 2.6 of [1], which proves inheritance under *contractive lattice-homomorphic retracts*. If the authors intended “complemented sublattice” to mean that narrower categorical notion, this packet is a scoped answer to the literal standard interpretation, not a solution of the narrower variant. It also does not settle the finite direct-sum, Fremlin-tensor-product, or extension questions.

5 Novelty check and review recommendation

Before promotion, the four lightweight run indexes were searched by exact arXiv id and semiprojectivity/permanence keywords. A bounded web/arXiv search through 11 August 2026 used the exact title, arXiv id, and combinations of “semiprojective Banach lattice”, “complemented sublattice”, and “ c_0 ”. It found the source preprint and summaries of it, but no later paper or note recording this counterexample. The elementary fact that $C[0, 1]$ contains a complemented sublattice isometric to c_0 is classical in spirit; the claimed new point is its combination with the source’s new semiprojectivity results. Novelty confidence is therefore bounded rather than exhaustive.

Human review should focus on the intended meaning of “complemented sublattice” in Question 5.14. Under bounded linear complementation, the proof is complete. Under lattice-homomorphic complementation, retain the construction only as a separation from that stronger notion.

References

- [1] T. Kania and M. Niwiński, *Semiprojective Banach lattices*, arXiv:2604.10624v1 (2026).